Neural Reservoir Surrogate: Technical Assessment Report

The objective of this assessment was to develop and evaluate a neural surrogate model for reservoir simulation using the PhysicsNeMo framework and the NORNE reservoir dataset. The task involved setting up the software environment, preprocessing reservoir simulation data, training an X-MeshGraphNet (X-MGN) model, performing inference on unseen test cases, and evaluating predictive performance through quantitative metrics and visual analysis. The NORNE dataset presents a challenging reservoir modeling problem due to the presence of geological faults represented by Non-Neighbor Connections (NNCs). These complex connectivity patterns make graph neural networks particularly suitable for the task. This report summarizes the complete workflow, including environment setup, data preparation, model configuration, training, inference, performance evaluation, failure analysis, and recommendations for future improvements.

1. Environment Setup

The PhysicsNeMo framework was installed in a dedicated Conda environment using Python 3.10 and the required project dependencies. Experiments were conducted on a workstation running Ubuntu 24.04 LTS (WSL2) with an NVIDIA RTX A4000 GPU (16 GB VRAM) and CUDA 12.1. The environment was successfully validated for data preprocessing, model training, and inference.

Issues Encountered and Solutions

Several setup and execution issues were encountered during the assessment and were successfully resolved. First, a Python version compatibility issue arose because PhysicsNeMo requires Python 3.10, whereas the host system was configured with Python 3.13. This was resolved by creating a dedicated Conda environment with Python 3.10. Second, recent versions of MLflow raised an exception when using filesystem-based tracking, which was addressed by enabling the `MLFLOW_ALLOW_FILE_STORE` environment variable. During preprocessing, auxiliary files within the NORNE dataset were incorrectly identified as simulation cases, this issue was resolved by creating a flat dataset layout containing only the required simulation files. In addition, timestep 17 of the NORNE_008 case was corrupted and automatically skipped by the preprocessing pipeline, leaving 59 usable cases for training and evaluation. Finally, an inference shape mismatch error was encountered during prediction reconstruction and was resolved through modifications to the inference pipeline, as described in Section 5.

2. Dataset & Preprocessing

The experiments were conducted on the NORNE reservoir simulation dataset, which consists of 60 realizations generated using Latin Hypercube Sampling (LHS). The reservoir model contains approximately 44,431 active cells on a 46 × 112 × 22 grid, with PRESSURE and SWAT used as prediction targets. Geological faults are represented through approximately 47 Non-Neighbor Connections (NNCs). The dataset was divided into training, validation, and test sets using an 80:10:10 ratio with a fixed random seed of 42, resulting in 48 training cases, 6 validation cases, and 6 test cases. The preprocessing pipeline converted the reservoir simulations into graph representations through graph construction, feature extraction, partitioning, and dataset splitting. A total of 3,720 graph samples were generated, producing 2,976 training samples, 372 validation samples, and 372 test samples.

Only a small number of configuration parameters were modified from the default settings. The `sim_dir` parameter was updated to point to the flattened NORNE dataset directory required for preprocessing, `random_seed` was set to 42 to ensure reproducibility, `num_epochs` was reduced from 1000 to 200 to fit the available compute budget, and `resume` was enabled to continue training from an existing checkpoint. Training converged and stopped automatically at epoch 105 due to early stopping, indicating that the reduced epoch limit did not negatively affect model performance.

3. Model Architecture

X-MeshGraphNet (X-MGN) is a graph neural network designed for learning physical processes on mesh-based domains. In the NORNE reservoir model, active grid cells are represented as graph nodes and fluid-flow connections as graph edges. The model operates autoregressively, predicting the reservoir state at the next timestep from the current state. The graph representation naturally incorporates geological faults through Non-Neighbor Connections (NNCs), making it well suited for faulted reservoir simulations. The model uses 13 node features consisting of static reservoir properties (PERMX, PORV, X, Y, Z) and dynamic variables (PRESSURE, SWAT, and WCID from the current and two previous timesteps). A single edge feature representing combined log-scaled transmissibility (TRANX, TRANY, TRANZ, and TRANNNC) is used to model fluid-flow connectivity. Two global features, timestep size (delta_t) and normalized simulation time, provide temporal information. The X-MGN model was configured with 5 message-passing layers, a hidden dimension of 128, and SiLU activation functions. PRESSURE was optimized using L2 (MSE) loss, while SWAT was optimized using L1 (MAE) loss, with equal loss weights of [1.0, 1.0].

4. Training

The X-MGN model was trained using the AdamW optimizer with a cosine annealing learning rate schedule. The learning rate decreased from 0.001 to 0.000001 during training, with weight decay set to 0.001. Early stopping was enabled with a patience of 20 validation checks and a minimum improvement threshold of 1×10^{-6}. Training was performed on 2,976 graph samples with 372 validation samples. Training was completed in two phases due to checkpoint resumption, covering a total of 105 epochs. The model achieved its best validation loss of 0.005454 at epoch 85 and stopped automatically at epoch 105 due to early stopping. Training required approximately 29 hours on an NVIDIA RTX A4000 GPU (about 15 minutes per epoch). The best checkpoint, `MeshGraphNet.0.85.mdlus`, represented a 43% improvement over the initial validation loss, while the close agreement between training and validation losses indicated no significant overfitting.

5. Inference Bug Fix

During inference, a shape mismatch error prevented successful reconstruction of full-grid predictions. Investigation revealed two issues in the prediction reassembly pipeline. First, the output array was incorrectly allocated using the size of a partition-level prediction array, while the permutation mapping corresponded to all active reservoir cells. This was resolved by allocating the output array according to the total number of active cells before applying the permutation mapping. Second, prediction outputs were stored as separate partition-level arrays, whereas the reordering procedure expected a single full-grid array. This issue was resolved by concatenating all partition outputs prior to reordering. Together, these fixes restored the inference pipeline and enabled successful generation of full-grid HDF5 prediction outputs and downstream visualizations.

6. Test Set Results

The trained X-MGN model was evaluated on six unseen test cases selected using a random seed of 42. Table 1 summarizes the prediction performance for PRESSURE and SWAT across the test set.

Case Study	Performance Category	PRESSURE RMSE (bar)	SWAT RMSE
NORNE_018	Best	9.14	0.01229
NORNE_008	Good	9.32	0.01230
NORNE_048	Good	9.35	0.01233
NORNE_041	Median	9.66	0.01237
NORNE_016	Poor	9.76	0.01246
NORNE_002	Worst	10.32	0.01247

Table 1. Representative Case Performance Based on PRESSURE and SWAT Prediction Accuracy.

6.1 Aggregate Performance Metrics Across all Test Cases

Aggregate performance statistics were computed across the six test cases to evaluate the overall predictive accuracy and robustness of the X-MGN model.

Variable	Metric	Mean	P50 (Median)	P10	P90	Worst
PRESSURE (bar)	RMSE	9.591	9.504	9.229	10.040	10.320
PRESSURE (bar)	MAE	4.736	4.748	4.662	4.797	4.803
SWAT (fraction)	RMSE	0.01237	0.01235	0.01230	0.01246	0.01247
SWAT (fraction)	MAE	0.003311	0.003294	0.003272	0.003367	0.003369

Table 2. Aggregate prediction performance statistics computed across all test cases

The model achieved a mean PRESSURE RMSE of 9.591 bar and a mean SWAT RMSE of 0.01237. The small differences between the P10, median, and P90 values indicate consistent performance across the test set, while the worst-case errors remain close to the average values, demonstrating good robustness and generalization capability.

6.2 Discussion

The X-MGN model achieved a mean PRESSURE MAE of 4.736 bar for a reservoir pressure range of approximately 87–549 bar, corresponding to a relative error of about 1.4%. This indicates that the model accurately captures the overall pressure dynamics of the reservoir despite the complex geological structure and fault interactions. For water saturation prediction, the model achieved a mean SWAT MAE of 0.003311 on a normalized scale of 0–1. The low error values demonstrate that the model successfully tracks the evolution of water saturation throughout the simulation period and across different reservoir realizations.

Performance was highly consistent across the six test cases. PRESSURE RMSE values ranged from 9.14 to 10.32 bar, while SWAT RMSE values varied only between 0.01229 and 0.01247. This narrow spread suggests strong generalization to unseen fault transmissibility configurations, despite training on only 48 reservoir realizations. Overall, the results indicate that X-MGN effectively learns the underlying reservoir flow dynamics and remains robust under geological uncertainty.

6.3 Autoregressive Error Drift

To evaluate long-term prediction stability, error metrics were analyzed at selected timesteps throughout the simulation horizon.

Timestep	PRESSURE RMSE (bar)	SWAT RMSE
3 (Early)	0.90	0.00118
8	7.04	0.01087
13	12.45	0.03529
23	3.32	0.00440
33	7.66	0.01484
43	5.74	0.00438
53	12.97	0.01104
63 (Final)	12.80	0.00650

Table 3. Temporal evolution of prediction errors for selected simulation timesteps

Early-timestep predictions are highly accurate, achieving a PRESSURE RMSE of only 0.90 bar and a SWAT RMSE of 0.00118 at timestep 3. As expected for an autoregressive forecasting model, prediction errors generally increase over time because inaccuracies from previous predictions propagate to subsequent timesteps. However, the error growth is not strictly monotonic. Error levels decrease around timestep 23 before increasing again at later stages of the simulation, indicating that the model successfully captures medium-term reservoir dynamics but experiences error accumulation over extended forecasting horizons. Despite this drift, the final timestep PRESSURE RMSE remains below 13 bar, demonstrating reasonable long-term stability for multi-step reservoir prediction.

7. Visualization

Two visualizations were generated for the worst-performing test case, **NORNE_002**, at timestep 23 to assess both the statistical and spatial accuracy of the X-MGN predictions.

7.1 Value Distribution (Histogram)

Figure 1 compares the distributions of TRUE, PRED, and DIFF values for PRESSURE and SWAT across all active reservoir cells. The predicted distributions closely overlap with the ground-truth distributions, indicating that the model accurately captures the statistical behavior of both variables. The error distributions are centered near zero, suggesting minimal systematic bias. At this timestep, the model achieved a PRESSURE RMSE of 3.66 bar and a SWAT RMSE of 0.0045, which are substantially lower than the overall test-set averages.

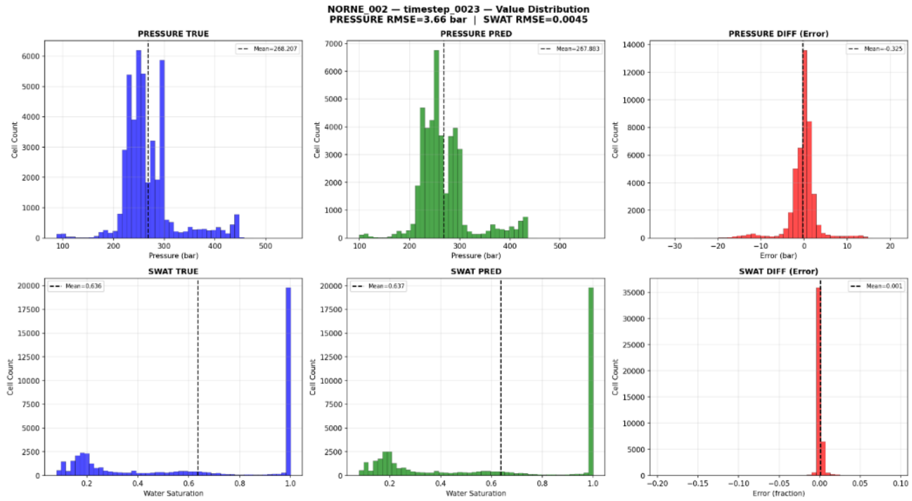

Figure 7.1. Histogram comparison of TRUE, PRED, and DIFF values for PRESSURE and SWAT in NORNE_002 at timestep 23.

7.2 Spatial Pattern Analysis (Top-Down View)

The spatial visualization (figure 2) presents top-down maps of PRESSURE and SWAT obtained by averaging values across the depth (Z) dimension. The predicted spatial patterns closely match the true reservoir behavior, including regions of high and low pressure and the evolution of water saturation fronts. Error maps reveal that the largest prediction discrepancies occur near reservoir boundaries and faulted regions represented by Non-Neighbor Connections (NNCs).

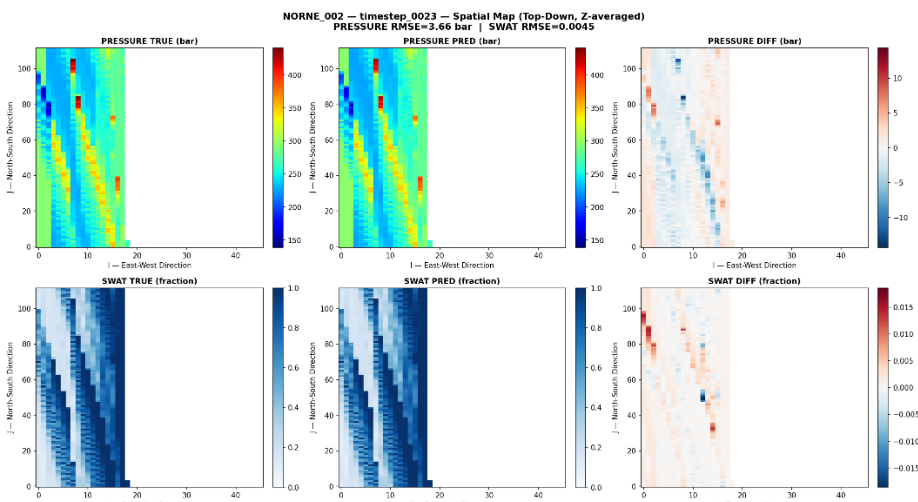

Figure 2. Top-down spatial comparison of TRUE, PRED, and DIFF maps for PRESSURE and SWAT in NORNE_002 at timestep 23.

The spatial visualizations show that the model accurately reproduces the major pressure gradients across the reservoir and successfully captures the evolution of water saturation fronts. Although slight spatial smearing is observed near fault connections, the overall agreement between the predicted and reference fields remains strong. The largest prediction errors are concentrated near reservoir boundaries and fault-affected regions, while the reservoir interior exhibits relatively small discrepancies. These observations indicate that the model preserves the key spatial flow patterns governing reservoir dynamics and achieves good spatial accuracy despite the presence of complex geological faults.

8. Failure Analysis

NORNE_002 was identified as the most challenging test case, producing the highest prediction errors among the six test cases (PRESSURE RMSE = 10.32 bar, SWAT RMSE = 0.01247). Although early-timestep predictions were highly accurate (PRESSURE RMSE = 0.90 bar at timestep 3), errors increased substantially at later timesteps, reaching 12.97 bar at timestep 53. Spatial visualizations indicate that the largest prediction errors occur near reservoir boundaries, fault-affected regions represented by Non-Neighbor Connections (NNCs), and at later timesteps where prediction errors accumulate.

Several factors may contribute to this behavior. NORNE_002 may contain fault transmissibility multipliers near the extremes of the Latin Hypercube Sampling design space, resulting in geological conditions that are underrepresented in the training set. In addition, the model must infer fault behavior indirectly from transmissibility values because fault multipliers are not provided as explicit features. However, the dominant failure mode appears to be autoregressive error accumulation, where small prediction errors propagate and compound over long forecasting horizons. This suggests that future improvements should focus on reducing long-term error drift through techniques such as scheduled sampling, teacher forcing, or multi-step training objectives.

9. FNO vs X-MGN for Faulted Reservoirs

Fourier Neural Operators (FNOs) perform global convolutions in the frequency domain and are designed primarily for structured, regular grids. While highly efficient for grid-based simulations, they face challenges when applied to faulted reservoirs such as Norne. The Norne reservoir contains approximately 47 geological faults represented as Non-Neighbor Connections (NNCs). These connections introduce fluid-flow pathways between spatially non-adjacent cells, breaking the regular grid structure assumed by FFT-based operators. Consequently, FNOs cannot naturally represent fault-controlled flow behavior and may smooth discontinuities across faults due to their frequency-domain formulation. In contrast, X-MGN models the reservoir as a graph, where both neighboring cells and fault-induced NNCs are represented as edges. This allows the network to directly capture complex connectivity patterns and geological discontinuities without requiring modifications to the underlying architecture. FNOs may outperform X-MGN for large-scale reservoirs with regular grid structures and no significant faulting, where their computational efficiency and global receptive field provide advantages. However, for faulted reservoirs such as Norne, the graph-based representation of X-MGN is fundamentally better suited because it explicitly incorporates the non-local flow connections created by faults. Therefore, X-MGN is not only a practical choice for the Norne dataset but also the more physically appropriate architecture for modeling fault-controlled reservoir flow.

10. Conclusion and Future Work

This assessment demonstrated the successful application of X-MeshGraphNet (X-MGN) for surrogate modeling of the NORNE reservoir dataset using the PhysicsNeMo framework. The model achieved a mean PRESSURE RMSE of 9.591 bar and a mean SWAT RMSE of 0.01237 on the held-out test set, while maintaining consistent performance across different fault transmissibility scenarios. Although autoregressive error accumulation was observed at later timesteps, the results show that X-MGN effectively captures reservoir dynamics and naturally represents fault-controlled flow through its graph-based architecture. Future improvements could include additional fine-tuning with a lower learning rate, increasing model capacity, incorporating fault multiplier values (MULTFLT) as explicit input features, and applying scheduled sampling to reduce long-term error drift. These enhancements could further improve prediction accuracy, robustness, and generalization to unseen reservoir scenarios.